Lab 9 Configure IPv4 and IPv6 Static and Default Routes

Topology

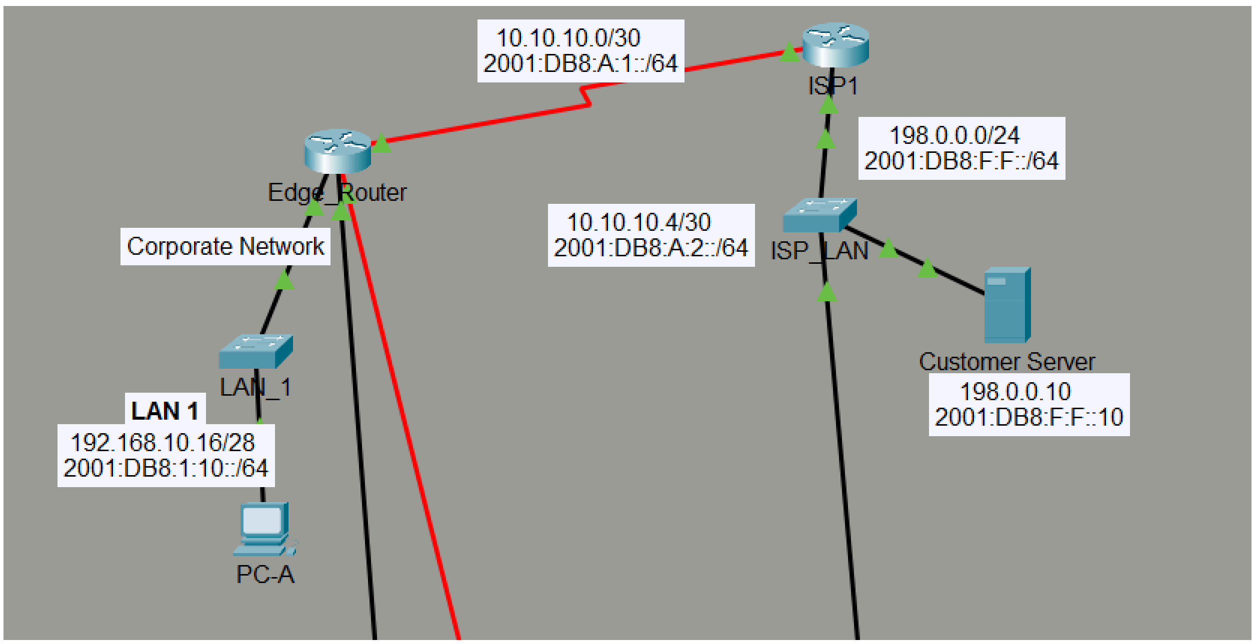

Table Address

Device	Interface	IP Address / Prefix
Edge_Router	S0/0/0	10.10.10.2/30
		2001:db8:a:1::2/64
	S0/0/1	10.10.10.6/30
		2001:db8:a:2::2/64
	G0/0	192.168.10.17/28
		2001:db8:1:10::1/64
	G0/1	192.168.11.33/27
		2001:db8:1:11::1/64
ISP1	S0/0/0	10.10.10.1/30
		2001:db8:a:1::1/64
	G0/0	198.0.0.1/24
		2001:db8:f:f::1/64
ISP2	S0/0/1	10.10.10.5/30
		2001:db8:a:2::1/64
	G0/0	198.0.0.2/24
		2001:db8:f:f::2/64
PC-A	NIC	192.168.10.19/28
		2001:db8:1:10::19/64
PC-B	NIC	192.168.11.4/27
		2001:db8:1:11::45
Customer Server	NIC	198.0.0.10
		2001:db8:f:f::10

EdgeRouter

```
Edge_router#show ip interface brief
Interface              IP-Address      OK? Method Status
GigabitEthernet0/0/0   192.168.10.17   YES manual up
GigabitEthernet0/0/1   10.10.10.2      YES manual up
GigabitEthernet0/0/2   unassigned      YES unset  down
GigabitEthernet0/0/3   unassigned      YES unset  down
Serial0/1/0            unassigned      YES unset  up
Serial0/1/1            unassigned      YES unset  up
Edge_router#
```

Default route

ip route 0.0.0.0 0.0.0.0 10.10.10.1
บอก edge router ให้ส่งไปที่ IP address 10.10.10.1 (ISP1)

EdgeRouter ping to ISP1

success

```
Edge_router>ping 10.10.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

ISP1

```
ISP1(config)#interface Serial0/1/0
ISP1(config-if)#no adress
                    ^
% Invalid input detected at '^' marker.

ISP1(config-if)#no address
                    ^
% Invalid input detected at '^' marker.

ISP1(config-if)#no ip address
ISP1(config-if)#exit
ISP1(config)#interface GigabitEthernet0/0/1
ISP1(config-if)#ip address 10.10.10.1 255.255.255.252
ISP1(config-if)#no ip address
ISP1(config-if)#ip address 10.10.10.1 255.255.255.252
ISP1(config-if)#no shutdown
ISP1(config-if)#end
ISP1#w
*Jul 23 10:14:22.177: %SYS-5-CONFIG_I: Configured from console by console
ISP1#write memory
Building configuration...
[OK]
ISP1#
*Jul 23 10:14:30.506: %SYS-6-PRIVCFG_ENCRYPT_SUCCESS: Successfully encrypted private config file
ISP1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   198.0.0.1       YES manual up                    up
GigabitEthernet0/0/1   10.10.10.1      YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Serial0/1/0            unassigned      YES manual up                    up
Serial0/1/1            209.165.200.225 YES manual up                    up
ISP1#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
ISP1#ping 192.168.10.17
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.17, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
ISP1#
```

Customer Server Pings

```
C:\Users\CY-VIP-Malware EArth>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CY-VIP-Malware EArth>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CY-VIP-Malware EArth>ping 192.168.10.19

Pinging 192.168.10.19 with 32 bytes of data:
Reply from 192.168.10.19: bytes=32 time<1ms TTL=126
Reply from 192.168.10.19: bytes=32 time<1ms TTL=126
Reply from 192.168.10.19: bytes=32 time=1ms TTL=126
Reply from 192.168.10.19: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.19:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\CY-VIP-Malware EArth>ping 192.168.10.17

Pinging 192.168.10.17 with 32 bytes of data:
Reply from 192.168.10.17: bytes=32 time<1ms TTL=254
Reply from 192.168.10.17: bytes=32 time<1ms TTL=254
Reply from 192.168.10.17: bytes=32 time<1ms TTL=254
Reply from 192.168.10.17: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CY-VIP-Malware EArth>
```

PC - A Ping to ISP1

```
C:\Users\computer>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```